

Pattern Recognition and Machine Learning

– Project –

General Project Structure

Each team must design and execute a project built around a real-world machine learning problem of their choice.

The project must contain the following mandatory components:

Comparative Study

Students must:

- a) Select one or more real-world datasets,
- b) Clearly define the machine learning problem,
- c) Perform exploratory data analysis,
- d) Apply and compare method(s) covered in the course,
- e) Implement the chosen baseline methods from scratch,
- f) Evaluate and interpret their performance.

Extension Component

In addition to baseline methods, each team must implement from scratch one of the following:

- A classical machine learning algorithm not covered in the course OR
- an existing extension/improvement of a covered algorithm

The method must already exist in literature or established practice and must be implemented from scratch (like in exercises). External libraries may only be used for validation/comparison, not as the primary implementation.

Students are encouraged to define their own project direction.

However, for those not sure where to start, here are some suggestions regarding algorithms:

- Support Vector Machines (SVM)
- Random Forests
- Gradient Boosting
- AdaBoost
- DBSCAN
- Weighted kNN
- Kernel PCA
- Adaptive KDE
- Regularized LDA

- k-means++

Also, some ideas about approaches:

- Focus on one dataset and compare many algorithms
- Focus on one algorithm across multiple datasets
- Study one advanced extension in depth
- Compare baseline methods and modern extensions
- Investigate algorithmic behavior under different data regimes

The exact project design is flexible, provided all mandatory components are satisfied.

Dataset Selection

Students may freely choose (real-world, appropriate in scope, ML-suitable) datasets, subject to instructor approval. Some examples of standard datasets:

- Wine dataset
- Breast Cancer Wisconsin dataset
- Dry Bean dataset
- Sonar dataset
- Titanic dataset
- Mall Customers dataset
- MNIST subset

Project Proposal

Before beginning project implementation, it is **not** mandatory – however it is advisable – to submit a short project proposal to the TA, containing:

- a) Selected dataset(s)
- b) Problem formulation
- c) Planned baseline algorithms
- d) Chosen independent extension method
- e) Reference source (paper, article, textbook, etc.)
- f) Preliminary implementation plan

If the project proposal was not sent for approval, it is not guaranteed that it is eligible for the maximum number of points allocated to the project.

Deliverables

Each team must submit:

- Full project report including but not limited to: introduction, problem description, dataset description, baseline, extension method(s), experimental setup, results, analysis of results, conclusions
- Jupyter notebook(s) and/or Python script(s) containing all implementations – via GitHub
- Presentation for oral defense

Example Project Direction

Example Title: Comparative Study of Classical and Extended Classification Methods on the Breast Cancer Wisconsin Dataset

Dataset: Breast Cancer Wisconsin Dataset.

Baseline methods: QDA, kNN.

Independent extension: weighted kNN / Support Vector Machine (implemented from scratch).

Possible research questions:

- Does the extension outperform baseline methods?
- Does PCA improve classification?
- How does model complexity compare to performance gain?

Outputs:

- Exploratory data analysis
- Method implementations
- Performance comparison
- Hyperparameter analysis
- Visualizations
- Discussion

This is only one example; students are encouraged to define their own project direction provided all core requirements are satisfied.